ALK fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ALK gene fusions from the msk_impact_50k_2026 study, covering 272 fusion events. 226/232 fusions with known frame status (97.4%) were in-frame. 261/271 fusions (96.3%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.00057792, statistically significant at alpha=0.05). No literature benchmark is configured for ALK.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 272 analyzed fusion events, identifying 32 distinct fusion partner genes. The most frequent partner was EML4, observed in 221 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions retained the domain, p=0.00057792 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00039996.

Domain disruption

Domain-disruption analysis was skipped: ALK has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

90% of 215 fusions with this gene as the 3' partner join at exon 20, against 0% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 224 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 972 aa (exon 17/18 boundary; label-permutation p=0.000999001). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 12 aa from the nearest configured domain boundary at 960 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=215) and "not_retained" (n=9). For Frame_status, retained: 186/190 (97.9%, 95% CI 94.7%-99.2%); not_retained: 3/4 (75.0%, 95% CI 30.1%-95.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 64.4019 vs 34.4444, p=0.0149559 (statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for ALK in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (8 row(s) in genomic_bin_recurrence, 14 row(s) in exact_position_recurrence, 5 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: ALK has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run (3 row(s) in retention_counter_intuitive_events); see the accompanying table.

Mutation cooccurrence

Mutation cooccurrence reported: ALK has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (119 row(s) in window_scan, 65 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	221	0.8125	0.33979834359489003	None	0.8867721063982746	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.6584926335411674	1
LRP1B	1	0.003676470588235294	0.33979834359489003	None	1.0	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3764140400620791	2
EMILIN1	1	0.003676470588235294	0.33979834359489003	None	0.9559412550066756	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.36815302537583083	3
CLTC	2	0.007352941176470588	0.33979834359489003	None	0.931909212283044	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3650256938357381	4
FN1	1	0.003676470588235294	0.33979834359489003	None	0.931909212283044	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.36364701736514987	5
NOTCH1	1	0.003676470588235294	0.33979834359489003	None	0.931909212283044	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.36364701736514987	6
NRP2	1	0.003676470588235294	0.33979834359489003	None	0.931909212283044	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.36364701736514987	7
STRN	7	0.025735294117647058	0.33979834359489003	None	0.8875166889185581	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3635954780578382	8
KIF5B	5	0.01838235294117647	0.33979834359489003	None	0.8968624833110814	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.36259046156525987	9
PLEKHA7	1	0.003676470588235294	0.33979834359489003	None	0.9158878504672897	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.36064301202469595	10
SPTBN1	4	0.014705882352941176	0.33979834359489003	None	0.8895193591455274	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35983494931363025	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RANBP2	3	0.011029411764705883	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3576426880633357	12
PICALM	3	0.011029411764705883	0.33979834359489003	None	0.8811748998664887	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3568916867282222	13
SQSTM1	2	0.007352941176470588	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35626401159274745	14
TPM1	2	0.007352941176470588	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35626401159274745	15
CTNNA1	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	16
DCTN1	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	17
DIAPH2	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	18
DYSF	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	19
GCC2	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	20
HMBOX1	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	21
PLEKHH2	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	22
PPP1CB	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	23
TFG	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TRAF3	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	25
TTC27	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	26
VCL	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	27
ZFPM2	1	0.003676470588235294	0.33979834359489003	None	0.8851802403204272	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35488533512215925	28
H6PD	1	0.003676470588235294	0.33979834359489003	None	0.7503337783711616	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3296016235066719	29
CTSE	1	0.003676470588235294	0.33979834359489003	None	0.39118825100133514	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2622618371248295	30
SOS1	1	0.003676470588235294	0.33979834359489003	None	None	0.6507870497447023	confidence_certainty, domain_retention, recurrence	0.2325095877687128	31
ASAP2	1	0.003676470588235294	0.33979834359489003	None	0.0	0.6507870497447023	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.18891404006207915	32

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
223	0	215	1	8	0.0	0.040178571428571425	1.396005508894838
516	0	215	2	7	0.0	0.001441383728379244	2.841220384951055
785	0	215	3	6	0.0	4.5449036480426615e-05	4.342475319387437
939	2	213	3	6	0.018779342723004695	0.0004361499679357765	3.360364155115966
972	2	213	4	5	0.011737089201877934	1.7840737389199315e-05	4.748587199430633
1023	10	205	4	5	0.06097560975609756	0.0010254772862659952	2.9890739545628864
1028	11	204	4	5	0.06740196078431372	0.001372173672346846	2.8625909176609534
1032	12	203	4	5	0.07389162561576355	0.0017951764400531472	2.7458928600797115
1035	13	202	4	5	0.08044554455445545	0.002303280356292077	2.637653196304051

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1040	14	201	4	5	0.08706467661691543	0.002905360704243936	2.536799941711491
1042	15	200	4	5	0.09375	0.003610320800972034	2.442454206431762
1045	17	198	4	5	0.10732323232323232	0.005364342117964091	2.2704835322361876
1047	18	197	4	5	0.11421319796954314	0.006430924027357664	2.191726620987009
1051	19	196	4	5	0.1211734693877551	0.007635344557662967	2.117171359902215
1052	21	194	4	5	0.13530927835051546	0.01049095972145548	1.979184780379653
1058	210	5	9	0	0.0	1.0	-0.0
1059	211	4	9	0	0.0	1.0	-0.0
1061	212	3	9	0	0.0	1.0	-0.0
1064	213	2	9	0	0.0	1.0	-0.0
1070	214	1	9	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

186	29
3	6

frame_domain_contingency_tables_by_tumor_type

stratum	table
ADNOS	[1, 0], [0, 0]
COAD	[3, 3], [0, 0]
CSCLC	[0, 1], [0, 0]
CUP	[1, 0], [0, 0]
GB	[1, 0], [0, 0]
HGSOC	[1, 0], [0, 0]
IDC	[1, 0], [0, 0]
IMT	[7, 0], [0, 0]
LNET	[0, 1], [0, 0]
LUAD	[152, 21], [1, 3]
LUAS	[1, 0], [0, 0]
LUSC	[1, 0], [0, 0]
MACR	[0, 1], [0, 0]
MPNST	[1, 0], [0, 0]

stratum	table
NSCLC	[1, 0], [0, 0]
NSCLCPD	[2, 0], [0, 0]
OSACA	[1, 0], [0, 0]
PAAD	[1, 1], [0, 1]
PDC	[1, 0], [0, 0]
PEMESO	[1, 0], [0, 0]
PRAD	[0, 0], [0, 1]
SAAD	[1, 0], [0, 0]
SARCL	[1, 0], [0, 0]
SARCNOS	[0, 0], [1, 0]
SDCA	[1, 0], [0, 0]
SKCM	[1, 0], [0, 0]
THHC	[0, 1], [0, 0]
THPD	[1, 0], [0, 0]
ULMS	[0, 0], [0, 1]
URCC	[3, 0], [0, 0]
USTUMP	[1, 0], [0, 0]
UTUC	[0, 0], [1, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	224	215	0	9	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
20	210	225	0.9333333333333333

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ASAP2	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CLTC	2	2	2	2
CTNNA1	1	1	1	1
CTSE	1	1	1	1
DCTN1	1	1	1	1
DIAPH2	1	1	1	1
DYSF	1	1	1	1
EMILIN1	1	1	1	1
EML4	221	221	220	182
FN1	1	1	1	1
GCC2	1	1	1	1
H6PD	1	1	1	1
HMBOX1	1	1	1	1
KIF5B	5	5	5	4
LRP1B	1	1	1	1
NOTCH1	1	1	1	1
NRP2	1	1	1	1
PICALM	3	3	3	1
PLEKHA7	1	1	1	1
PLEKHH2	1	1	1	1
PPP1CB	1	1	1	1
RANBP2	3	3	3	3
SOS1	1	1	1	1
SPTBN1	4	4	4	4
SQSTM1	2	2	2	1
STRN	7	7	7	4
TFG	1	1	1	1
TPM1	2	2	2	1
TRAF3	1	1	1	1
TTC27	1	1	1	1
VCL	1	1	1	1
ZFPM2	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
2	29447000	29448000	97	EVT-P-0000474-T01-IM3-107, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, EVT-P-0002024-T01-IM3-163, ... (97 total)
2	29446000	29447000	74	EVT-P-0000557-T01-IM3-113, EVT-P-0002217-T02-IM3-164, EVT-P-0003446-T01-IM5-99, EVT-P-0004516-T02-IM5-171, EVT-P-0007125-T01-IM5-151, ... (74 total)
2	29448000	29449000	44	EVT-P-0001766-T04-IM6-222, EVT-P-0001811-T01-IM3-154, EVT-P-0002408-T02-IM5-78, EVT-P-0003206-T01-IM5-90, EVT-P-0003297-T01-IM5-91, ... (44 total)
2	29449000	29450000	3	EVT-P-0005711-T02-IM6-294, EVT-P-0062994-T02-IM7-188, EVT-P-0067512-T01-IM7-297
2	29450000	29451000	3	EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0031824-T01-IM6-285
2	29462000	29463000	1	EVT-P-0035712-T01-IM6-172
2	29543000	29544000	1	EVT-P-0024679-T01-IM6-39
2	30142000	30143000	1	EVT-P-0006617-T01-IM5-187

exact_position_recurrence

chromosome	position	n_events	event_ids
2	29446597	3	EVT-P-0007125-T01-IM5-151, EVT-P-0013453-T01-IM5-54, EVT-P-0043041-T01-IM6-227
2	29446465	2	EVT-P-0009353-T01-IM5-162, EVT-P-0025388-T01-IM6-190
2	29446514	2	EVT-P-0034569-T02-IM6-143, EVT-P-0045363-T01-IM6-318
2	29446526	2	EVT-P-0004516-T02-IM5-171, EVT-P-0015961-T01-IM6-65
2	29446534	2	EVT-P-0003446-T01-IM5-99, EVT-P-0064342-T01-IM7-274
2	29446609	2	EVT-P-0017978-T01-IM6-73, EVT-P-0047355-T01-IM6-245
2	29447002	2	EVT-P-0000474-T01-IM3-107, EVT-P-0021693-T01-IM6-100
2	29447079	2	EVT-P-0015152-T02-IM6-61, EVT-P-0035085-T03-IM6-200
2	29447199	2	EVT-P-0002613-T01-IM3-88, EVT-P-0062985-T02-IM7-270
2	29447321	2	EVT-P-0025095-T01-IM6-198, EVT-P-0040576-T01-IM6-219
2	29447688	2	EVT-P-0040600-T01-IM6-220, EVT-P-0053383-T01-IM6-320
2	29447764	2	EVT-P-0005305-T01-IM5-111, EVT-P-0069535-T01-IM7-287
2	29448101	2	EVT-P-0019972-T01-IM6-85, EVT-P-0043363-T01-IM6-230
2	29448365	2	EVT-P-0003206-T01-IM5-90, EVT-P-0033066-T01-IM6-314

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
1058	194	180	1913	False
1023	8	8	1091	False
939	2	2	184	False
1045	2	1	0	False
1052	2	2	1	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	20	193	215	0.8976744186046511	0.0029075471361552593	None
three_prime	19	20	215	0.09302325581395349	0.0020046844577266795	26.254162194793984
five_prime	19	5	9	0.5555555555555556	0.0027950323342842208	10.671745810345206
three_prime	17	2	215	0.009302325581395349	0.001797492566476352	1.2379678104415628
five_prime	1	1	9	0.1111111111111111	0.2797351236908148	0.02327006111421805
five_prime	7	1	9	0.1111111111111111	0.003400142427273588	1.5201603048999828
five_prime	13	1	9	0.1111111111111111	0.008416656032645757	1.135197602647274
five_prime	17	1	9	0.1111111111111111	0.0008205347746205027	2.1330855237445734

mechanistic_interpretation tables

retention_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0026650-T02-IM6-114	P-0026650-T02-IM6	EML4	1058	lost
EVT-P-0018541-T01-IM6-174	P-0018541-T01-IM6	LRP1B	972	lost
EVT-P-0068865-T02-IM7-179	P-0068865-T02-IM7	RANBP2	1058	lost

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
772	972	200	4	220	2	213	0.03286384976525822	0.008288411809910093	2.0815286791325693	EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0031824-T01-IM6-285, EVT-P-0035712-T01-IM6-172
785	985	200	4	220	2	213	0.03286384976525822	0.008288411809910093	2.0815286791325693	EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0031824-T01-IM6-285, EVT-P-0035712-T01-IM6-172
823	1023	200	11	213	10	205	0.3902439024390244	0.3697928029452996	0.43204154550664126	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0023962-T01-IM6-109, ... (11 total)
828	1028	200	12	212	11	204	0.43137254901960786	0.39642127606028704	0.4018430448335435	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (12 total)
832	1032	200	13	211	12	203	0.4729064039408867	0.42204490113319926	0.3746413422207224	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (13 total)
835	1035	200	14	210	13	202	0.5148514851485149	0.4466970143550061	0.34998695026708926	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (14 total)
840	1040	200	15	209	14	201	0.5572139303482587	0.4704099994540772	0.3275234550833397	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (15 total)
842	1042	200	16	208	15	200	0.6	0.49321531048237055	0.30696345041611944	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (16 total)
845	1045	200	18	206	17	198	0.6868686868686869	0.5362242118084903	0.27065358047355936	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (18 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
847	1047	200	19	205	18	197	0.7309644670050761	0.5564862608071484	0.25454555357419484	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (19 total)
851	1051	200	20	204	19	196	0.7755102040816326	0.5759575956985422	0.23960948989630798	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (20 total)
852	1052	200	22	202	21	194	0.865979381443299	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (22 total)
858	1058	200	216	8	210	5	21.0	0.0022953048547373106	2.6391596247508766	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (216 total)
859	1059	200	217	7	211	4	26.375	0.0014646533971695624	2.834265136747466	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (217 total)
861	1061	200	218	6	212	3	35.333333333333336	0.0008544591984823537	3.0683086705641145	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
864	1064	200	219	5	213	2	53.25	0.0004361499679357765	3.360364155115966	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (219 total)
870	1070	200	220	4	214	1	107.0	0.0001780944144436627	3.7493497010552006	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (220 total)
923	1023	100	11	213	10	205	0.3902439024390244	0.3697928029452996	0.43204154550664126	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0023962-T01-IM6-109, ... (11 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
928	1028	100	12	212	11	204	0.43137254901960786	0.39642127606028704	0.4018430448335435	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (12 total)
932	1032	100	13	211	12	203	0.4729064039408867	0.42204490113319926	0.3746413422207224	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (13 total)
935	1035	100	14	210	13	202	0.5148514851485149	0.4466970143550061	0.34998695026708926	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (14 total)
939	1039	100	14	210	13	202	0.5148514851485149	0.4466970143550061	0.34998695026708926	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (14 total)
940	1040	100	13	211	12	203	0.4729064039408867	0.42204490113319926	0.3746413422207224	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (13 total)
942	1042	100	14	210	13	202	0.5148514851485149	0.4466970143550061	0.34998695026708926	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (14 total)
945	1045	100	16	208	15	200	0.6	0.49321531048237055	0.30696345041611944	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (16 total)
947	1047	100	17	207	16	199	0.6432160804020101	0.515143494163422	0.2880717805839425	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (17 total)
951	1051	100	18	206	17	198	0.6868686868686869	0.5362242118084903	0.27065358047355936	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (18 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
952	1052	100	20	204	19	196	0.7755102040816326	0.5759575956985422	0.23960948989630798	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (20 total)
958	1058	100	214	10	208	7	14.857142857142858	0.004717859407126972	2.3262550052941537	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (214 total)
959	1059	100	215	9	209	6	17.416666666666668	0.0033720752626955042	2.4721027407470046	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (215 total)
961	1061	100	216	8	210	5	21.0	0.0022953048547373106	2.6391596247508766	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (216 total)
964	1064	100	217	7	211	4	26.375	0.0014646533971695624	2.834265136747466	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (217 total)
970	1070	100	218	6	212	3	35.333333333333336	0.0008544591984823537	3.0683086705641145	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
971	1071	100	219	5	213	2	53.25	0.0004361499679357765	3.360364155115966	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (219 total)
972	1072	100	219	5	213	2	53.25	0.0004361499679357765	3.360364155115966	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (219 total)
972	1172	200	219	5	213	2	53.25	0.0004361499679357765	3.360364155115966	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (219 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
973	1023	50	8	216	8	207	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0023962-T01-IM6-109, EVT-P-0034044-T01-IM6-298, EVT-P-0044526-T02-IM6-289, ... (8 total)
978	1028	50	9	215	9	206	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0023962-T01-IM6-109, EVT-P-0034044-T01-IM6-298, ... (9 total)
982	1032	50	10	214	10	205	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (10 total)
985	1035	50	11	213	11	204	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (11 total)
990	1040	50	12	212	12	203	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (12 total)
992	1042	50	13	211	13	202	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (13 total)
995	1045	50	15	209	15	200	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (15 total)
997	1047	50	16	208	16	199	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (16 total)
998	1023	25	8	216	8	207	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0023962-T01-IM6-109, EVT-P-0034044-T01-IM6-298, EVT-P-0044526-T02-IM6-289, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1001	1051	50	17	207	17	198	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (17 total)
1002	1052	50	19	205	19	196	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (19 total)
1003	1028	25	9	215	9	206	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0023962-T01-IM6-109, EVT-P-0034044-T01-IM6-298, ... (9 total)
1007	1032	25	10	214	10	205	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (10 total)
1008	1058	50	213	11	208	7	23.771428571428572	0.0003576864781278212	3.446497477478851	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (213 total)
1009	1059	50	214	10	209	6	27.866666666666667	0.0002319122590220765	3.634676293686621	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (214 total)
1010	1035	25	11	213	11	204	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (11 total)
1011	1061	50	215	9	210	5	33.6	0.0001417640388209254	3.848433922342923	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (215 total)
1014	1064	50	216	8	211	4	42.2	8.023430122331438e-05	4.095639925390925	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (216 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1015	1040	25	12	212	12	203	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (12 total)
1017	1042	25	13	211	13	202	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (13 total)
1020	1045	25	15	209	15	200	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (15 total)
1020	1070	50	217	7	212	3	56.53333333333333	4.086693356607513e-05	4.388627948177679	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (217 total)
1021	1071	50	218	6	213	2	85.2	1.7840737389199315e-05	4.748587199430633	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
1022	1047	25	16	208	16	199	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (16 total)
1023	1048	25	16	208	16	199	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (16 total)
1023	1073	50	218	6	213	2	85.2	1.7840737389199315e-05	4.748587199430633	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
1023	1123	100	218	6	213	2	85.2	1.7840737389199315e-05	4.748587199430633	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1023	1223	200	218	6	213	2	85.2	1.7840737389199315e-05	4.748587199430633	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
1026	1051	25	9	215	9	206	None	1.0	-0.0	EVT-P-0003206-T01-IM5-90, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0032518-T01-IM6-138, EVT-P-0033066-T01-IM6-314, ... (9 total)
1027	1052	25	11	213	11	204	None	1.0	-0.0	EVT-P-0003206-T01-IM5-90, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0032518-T01-IM6-138, EVT-P-0033066-T01-IM6-314, ... (11 total)
1028	1053	25	11	213	11	204	None	1.0	-0.0	EVT-P-0003206-T01-IM5-90, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0032518-T01-IM6-138, EVT-P-0033066-T01-IM6-314, ... (11 total)
1028	1078	50	210	14	205	10	16.4	0.0010254772862659952	2.9890739545628864	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (210 total)
1028	1128	100	210	14	205	10	16.4	0.0010254772862659952	2.9890739545628864	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (210 total)
1028	1228	200	210	14	205	10	16.4	0.0010254772862659952	2.9890739545628864	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (210 total)
1032	1057	25	10	214	10	205	None	1.0	-0.0	EVT-P-0003206-T01-IM5-90, EVT-P-0019513-T01-IM6-82, EVT-P-0032518-T01-IM6-138, EVT-P-0033066-T01-IM6-314, EVT-P-0043594-T01-IM6-231, ... (10 total)
1032	1082	50	209	15	204	11	14.836363636363636	0.001372173672346846	2.8625909176609534	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (209 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1032	1132	100	209	15	204	11	14.836363636363636	0.001372173672346846	2.8625909176609534	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (209 total)
1032	1232	200	209	15	204	11	14.836363636363636	0.001372173672346846	2.8625909176609534	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (209 total)
1033	1058	25	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)
1034	1059	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)
1035	1060	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)
1035	1085	50	208	16	203	12	13.533333333333333	0.0017951764400531472	2.7458928600797115	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (208 total)
1035	1135	100	208	16	203	12	13.533333333333333	0.0017951764400531472	2.7458928600797115	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (208 total)
1035	1235	200	208	16	203	12	13.533333333333333	0.0017951764400531472	2.7458928600797115	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (208 total)
1036	1061	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1039	1064	25	205	19	200	15	10.666666666666666	0.003610320800972034	2.442454206431762	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (205 total)
1040	1065	25	205	19	200	15	10.666666666666666	0.003610320800972034	2.442454206431762	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (205 total)
1040	1090	50	207	17	202	13	12.430769230769231	0.002303280356292077	2.637653196304051	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (207 total)
1040	1140	100	207	17	202	13	12.430769230769231	0.002303280356292077	2.637653196304051	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (207 total)
1040	1240	200	207	17	202	13	12.430769230769231	0.002303280356292077	2.637653196304051	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (207 total)
1042	1067	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)
1042	1092	50	206	18	201	14	11.485714285714286	0.002905360704243936	2.536799941711491	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (206 total)
1042	1142	100	206	18	201	14	11.485714285714286	0.002905360704243936	2.536799941711491	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (206 total)
1042	1242	200	206	18	201	14	11.485714285714286	0.002905360704243936	2.536799941711491	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (206 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1045	1070	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)
1045	1095	50	205	19	200	15	10.666666666666666	0.003610320800972034	2.442454206431762	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (205 total)
1045	1145	100	205	19	200	15	10.666666666666666	0.003610320800972034	2.442454206431762	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (205 total)
1045	1245	200	205	19	200	15	10.666666666666666	0.003610320800972034	2.442454206431762	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (205 total)
1046	1071	25	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)
1047	1072	25	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)
1047	1097	50	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)
1047	1147	100	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)
1047	1247	200	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1051	1076	25	202	22	197	18	8.755555555555556	0.006430924027357664	2.191726620987009	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (202 total)
1051	1101	50	202	22	197	18	8.755555555555556	0.006430924027357664	2.191726620987009	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (202 total)
1051	1151	100	202	22	197	18	8.755555555555556	0.006430924027357664	2.191726620987009	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (202 total)
1051	1251	200	202	22	197	18	8.755555555555556	0.006430924027357664	2.191726620987009	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (202 total)
1052	1077	25	201	23	196	19	8.25263157894737	0.007635344557662967	2.117171359902215	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (201 total)
1052	1102	50	201	23	196	19	8.25263157894737	0.007635344557662967	2.117171359902215	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (201 total)
1052	1152	100	201	23	196	19	8.25263157894737	0.007635344557662967	2.117171359902215	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (201 total)
1052	1252	200	201	23	196	19	8.25263157894737	0.007635344557662967	2.117171359902215	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (201 total)
1058	1083	25	199	25	194	21	7.390476190476191	0.01049095972145548	1.979184780379653	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (199 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1058	1108	50	199	25	194	21	7.390476190476191	0.01049095972145548	1.979184780379653	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (199 total)
1058	1158	100	199	25	194	21	7.390476190476191	0.01049095972145548	1.979184780379653	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (199 total)
1058	1258	200	199	25	194	21	7.390476190476191	0.01049095972145548	1.979184780379653	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (199 total)
1059	1084	25	5	219	5	210	None	1.0	-0.0	EVT-P-0026407-T01-IM6-170, EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1059	1109	50	5	219	5	210	None	1.0	-0.0	EVT-P-0026407-T01-IM6-170, EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1059	1159	100	5	219	5	210	None	1.0	-0.0	EVT-P-0026407-T01-IM6-170, EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1059	1259	200	5	219	5	210	None	1.0	-0.0	EVT-P-0026407-T01-IM6-170, EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1061	1086	25	4	220	4	211	None	1.0	-0.0	EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1061	1111	50	4	220	4	211	None	1.0	-0.0	EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1061	1161	100	4	220	4	211	None	1.0	-0.0	EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1061	1261	200	4	220	4	211	None	1.0	-0.0	EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1021	1071	50	218	6	213	2	85.2	1.7840737389199315e-05	4.748587199430633	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
1020	1070	50	217	7	212	3	56.53333333333333	4.086693356607513e-05	4.388627948177679	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (217 total)
1014	1064	50	216	8	211	4	42.2	8.023430122331438e-05	4.095639925390925	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (216 total)
1011	1061	50	215	9	210	5	33.6	0.0001417640388209254	3.848433922342923	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (215 total)
870	1070	200	220	4	214	1	107.0	0.0001780944144436627	3.7493497010552006	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (220 total)
1009	1059	50	214	10	209	6	27.866666666666667	0.0002319122590220765	3.634676293686621	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (214 total)
1008	1058	50	213	11	208	7	23.771428571428572	0.0003576864781278212	3.446497477478851	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (213 total)
971	1071	100	219	5	213	2	53.25	0.0004361499679357765	3.360364155115966	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (219 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
864	1064	200	219	5	213	2	53.25	0.0004361499679357765	3.360364155115966	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (219 total)
970	1070	100	218	6	212	3	35.333333333333336	0.0008544591984823537	3.0683086705641145	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
861	1061	200	218	6	212	3	35.333333333333336	0.0008544591984823537	3.0683086705641145	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (218 total)
1028	1078	50	210	14	205	10	16.4	0.0010254772862659952	2.9890739545628864	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (210 total)
1032	1082	50	209	15	204	11	14.836363636363636	0.001372173672346846	2.8625909176609534	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (209 total)
964	1064	100	217	7	211	4	26.375	0.0014646533971695624	2.834265136747466	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (217 total)
859	1059	200	217	7	211	4	26.375	0.0014646533971695624	2.834265136747466	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (217 total)
1035	1085	50	208	16	203	12	13.533333333333333	0.0017951764400531472	2.7458928600797115	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (208 total)
961	1061	100	216	8	210	5	21.0	0.0022953048547373106	2.6391596247508766	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (216 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
858	1058	200	216	8	210	5	21.0	0.0022953048547373106	2.6391596247508766	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (216 total)
1040	1090	50	207	17	202	13	12.430769230769231	0.002303280356292077	2.637653196304051	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (207 total)
1042	1092	50	206	18	201	14	11.485714285714286	0.002905360704243936	2.536799941711491	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (206 total)
959	1059	100	215	9	209	6	17.416666666666668	0.0033720752626955042	2.4721027407470046	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (215 total)
1039	1064	25	205	19	200	15	10.666666666666666	0.003610320800972034	2.442454206431762	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (205 total)
1045	1095	50	205	19	200	15	10.666666666666666	0.003610320800972034	2.442454206431762	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (205 total)
1034	1059	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)
1036	1061	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)
1042	1067	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1045	1070	25	204	20	199	16	9.95	0.004427042864254589	2.3538862730021513	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (204 total)
958	1058	100	214	10	208	7	14.857142857142858	0.004717859407126972	2.3262550052941537	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (214 total)
1033	1058	25	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)
1046	1071	25	203	21	198	17	9.31764705882353	0.005364342117964091	2.2704835322361876	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (203 total)
1051	1076	25	202	22	197	18	8.755555555555556	0.006430924027357664	2.191726620987009	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (202 total)
1052	1077	25	201	23	196	19	8.25263157894737	0.007635344557662967	2.117171359902215	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (201 total)
772	972	200	4	220	2	213	0.03286384976525822	0.008288411809910093	2.0815286791325693	EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0031824-T01-IM6-285, EVT-P-0035712-T01-IM6-172
1058	1083	25	199	25	194	21	7.390476190476191	0.01049095972145548	1.979184780379653	EVT-P-0000474-T01-IM3-107, EVT-P-0000557-T01-IM3-113, EVT-P-0000732-T01-IM3-105, EVT-P-0001532-T01-IM3-145, EVT-P-0001737-T01-IM3-321, ... (199 total)
923	1023	100	11	213	10	205	0.3902439024390244	0.3697928029452996	0.43204154550664126	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0023962-T01-IM6-109, ... (11 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
928	1028	100	12	212	11	204	0.43137254901960786	0.39642127606028704	0.4018430448335435	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (12 total)
932	1032	100	13	211	12	203	0.4729064039408867	0.42204490113319926	0.3746413422207224	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (13 total)
940	1040	100	13	211	12	203	0.4729064039408867	0.42204490113319926	0.3746413422207224	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (13 total)
935	1035	100	14	210	13	202	0.5148514851485149	0.4466970143550061	0.34998695026708926	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (14 total)
942	1042	100	14	210	13	202	0.5148514851485149	0.4466970143550061	0.34998695026708926	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (14 total)
840	1040	200	15	209	14	201	0.5572139303482587	0.4704099994540772	0.3275234550833397	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (15 total)
945	1045	100	16	208	15	200	0.6	0.49321531048237055	0.30696345041611944	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (16 total)
842	1042	200	16	208	15	200	0.6	0.49321531048237055	0.30696345041611944	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (16 total)
947	1047	100	17	207	16	199	0.6432160804020101	0.515143494163422	0.2880717805839425	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (17 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
951	1051	100	18	206	17	198	0.6868686868686869	0.5362242118084903	0.27065358047355936	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (18 total)
845	1045	200	18	206	17	198	0.6868686868686869	0.5362242118084903	0.27065358047355936	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (18 total)
847	1047	200	19	205	18	197	0.7309644670050761	0.5564862608071484	0.25454555357419484	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (19 total)
952	1052	100	20	204	19	196	0.7755102040816326	0.5759575956985422	0.23960948989630798	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018541-T01-IM6-174, EVT-P-0018969-T01-IM6-80, ... (20 total)
851	1051	200	20	204	19	196	0.7755102040816326	0.5759575956985422	0.23960948989630798	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (20 total)
998	1023	25	8	216	8	207	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0023962-T01-IM6-109, EVT-P-0034044-T01-IM6-298, EVT-P-0044526-T02-IM6-289, ... (8 total)
1003	1028	25	9	215	9	206	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0023962-T01-IM6-109, EVT-P-0034044-T01-IM6-298, ... (9 total)
1007	1032	25	10	214	10	205	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (10 total)
1010	1035	25	11	213	11	204	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (11 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1015	1040	25	12	212	12	203	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (12 total)
1017	1042	25	13	211	13	202	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0023962-T01-IM6-109, ... (13 total)
1020	1045	25	15	209	15	200	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (15 total)
1022	1047	25	16	208	16	199	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (16 total)
1026	1051	25	9	215	9	206	None	1.0	-0.0	EVT-P-0003206-T01-IM5-90, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0032518-T01-IM6-138, EVT-P-0033066-T01-IM6-314, ... (9 total)
1027	1052	25	11	213	11	204	None	1.0	-0.0	EVT-P-0003206-T01-IM5-90, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, EVT-P-0032518-T01-IM6-138, EVT-P-0033066-T01-IM6-314, ... (11 total)
1032	1057	25	10	214	10	205	None	1.0	-0.0	EVT-P-0003206-T01-IM5-90, EVT-P-0019513-T01-IM6-82, EVT-P-0032518-T01-IM6-138, EVT-P-0033066-T01-IM6-314, EVT-P-0043594-T01-IM6-231, ... (10 total)
1059	1084	25	5	219	5	210	None	1.0	-0.0	EVT-P-0026407-T01-IM6-170, EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299
1061	1086	25	4	220	4	211	None	1.0	-0.0	EVT-P-0034450-T01-IM6-142, EVT-P-0049318-T01-IM6-248, EVT-P-0053901-T01-IM6-315, EVT-P-0055667-T02-IM6-299

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1001	1051	50	17	207	17	198	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (17 total)
1002	1052	50	19	205	19	196	None	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0018969-T01-IM6-80, EVT-P-0019513-T01-IM6-82, ... (19 total)
852	1052	200	22	202	21	194	0.865979381443299	1.0	-0.0	EVT-P-0001766-T04-IM6-222, EVT-P-0003206-T01-IM5-90, EVT-P-0005711-T02-IM6-294, EVT-P-0005759-T02-IM6-192, EVT-P-0018541-T01-IM6-174, ... (22 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024679-T01-IM6-39	P-0024679-T01-IM6	P-0024679	Prostate Cancer	PRAD	ALK-CTSE	CTSE	unknown	five_prime	29543594	7	516	True	515	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ALK (NM_004304) - CTSE (NM_001910) Rearrangement: t(1;2)(q32.1;p23.3)(chr1:g.206318648; chr2:g.29543594) | Antisense Fusion	NA
EVT-P-0020387-T01-IM6-40	P-0020387-T01-IM6	P-0020387	Soft Tissue Sarcoma	IMT	ALK-DCTN1	DCTN1	in-frame	three_prime	29446896	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			DCTN1 (NM_004082) - ALK (NM_004304) rearrangement: c.3196+238:DCTN1_c.3173-502:ALKdel | Protein Fusion: in frame {DCTN1:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006776-T01-IM5-44	P-0006776-T01-IM5	P-0006776	Non-Small Cell Lung Cancer	LUAD	ALK-DYSF	DYSF	unknown	five_prime	29447418	19	1058	True	1057	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ALK (NM_004304) - DYSF (NM_001130987) rearrangement: c.3172+909:ALK_c.3402+7348:DYSFdup | Antisense fusion	NA
EVT-P-0020825-T03-IM6-45	P-0020825-T03-IM6	P-0020825	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446767	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29):c.208+4890:EML4_c.3173-373:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011018-T01-IM5-46	P-0011018-T01-IM5	P-0011018	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448314	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011203-T01-IM5-47	P-0011203-T01-IM5	P-0011203	Salivary Gland Cancer	SDCA	ALK-EML4	EML4	in-frame	three_prime	29447105	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011203-T02-IM5-48	P-0011203-T02-IM5	P-0011203	Salivary Gland Cancer	SDCA	ALK-EML4	EML4	in-frame	three_prime	29447096	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6805:EML4_c.3173-702:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011203-T03-IM6-49	P-0011203-T03-IM6	P-0011203	Salivary Gland Cancer	SDCA	ALK-EML4	EML4	in-frame	three_prime	29447055	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 5 fused to ALK exon 20): c.494-6851:EML4_c.3173-661:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012679-T01-IM5-50	P-0012679-T01-IM5	P-0012679	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447466	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012778-T01-IM5-51	P-0012778-T01-IM5	P-0012778	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447442	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320:EML4_c.3172+885:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012778-T02-IM6-52	P-0012778-T02-IM6	P-0012778	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447442	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320:EML4_c.3172+885:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013202-T01-IM5-53	P-0013202-T01-IM5	P-0013202	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447178	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+2565:EML4_c.3173-784:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013453-T01-IM5-54	P-0013453-T01-IM5	P-0013453	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446597	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+551:EML4_c.3173-203:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013864-T01-IM5-55	P-0013864-T01-IM5	P-0013864	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448209	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1489+658:EML4_c.3172+118:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013869-T01-IM5-56	P-0013869-T01-IM5	P-0013869	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446712	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014270-T01-IM6-57	P-0014270-T01-IM6	P-0014270	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446679	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294:EML4_c.3173-285:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014310-T01-IM6-58	P-0014310-T01-IM6	P-0014310	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447544	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20- 29): c.667+6128:EML4_c.3172+783:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014385-T01-IM6-59	P-0014385-T01-IM6	P-0014385	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447676	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1890:EML4_c.3172+651:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015110-T01-IM6-60	P-0015110-T01-IM6	P-0015110	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446518	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668-1726:EML4_c.3173-124:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015152-T02-IM6-61	P-0015152-T02-IM6	P-0015152	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447079	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused in-frame to ALK exons 20-29): c.2242+127:EML4_c.3173-685:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015197-T01-IM6-62	P-0015197-T01-IM6	P-0015197	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448009	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6 fused to ALK exon 20): c.668-2108:EML4_c.3172+318:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015434-T02-IM6-63	P-0015434-T02-IM6	P-0015434	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448197	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015921-T03-IM6-64	P-0015921-T03-IM6	P-0015921	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446809	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.667+516:EML4_c.3173-415:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015961-T01-IM6-65	P-0015961-T01-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6fused to ALK exon 20): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015961-T02-IM6-66	P-0015961-T02-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015961-T03-IM6-67	P-0015961-T03-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAS	ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016280-T01-IM6-68	P-0016280-T01-IM6	P-0016280	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447905	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016580-T01-IM6-69	P-0016580-T01-IM6	P-0016580	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447579	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002266-T01-IM3-70	P-0002266-T01-IM3	P-0002266	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	five_prime	29447288	19	1058	True	1057	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ALK (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173-889:ALK_c.208+1022:EML4inv | Protein fusion: in frame (ALK-EML4)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016960-T01-IM6-71	P-0016960-T01-IM6	P-0016960	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447838	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017592-T01-IM5-72	P-0017592-T01-IM5	P-0017592	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447548	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29) : c.494-5963:EML4_c.3172+779:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017978-T01-IM6-73	P-0017978-T01-IM6	P-0017978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446609	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017978-T02-IM6-74	P-0017978-T02-IM6	P-0017978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446609	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017978-T03-IM6-75	P-0017978-T03-IM6	P-0017978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446609	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018242-T01-IM6-76	P-0018242-T01-IM6	P-0018242	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448007	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused in-frame with ALK exons 20-29): c.668-7560:EML4_c.3172+320:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018280-T01-IM6-77	P-0018280-T01-IM6	P-0018280	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447187	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 6 fused with ALK exon 20): c.494-6792:EML4_c.3173-793:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002408-T02-IM5-78	P-0002408-T02-IM5	P-0002408	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448053	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fuses with ALK exons 20-29) : c.2057-115:EML4_c.3172+274:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018969-T01-IM6-80	P-0018969-T01-IM6	P-0018969	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29448416	19	1028	False	1028	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-760:EML4_c.3083:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019491-T01-IM6-81	P-0019491-T01-IM6	P-0019491	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446596	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1490-486:EML4_c.3173-202:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019513-T01-IM6-82	P-0019513-T01-IM6	P-0019513	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29448403	19	1032	False	1032	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1489+2278:EML4_c.3096:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019533-T01-IM6-83	P-0019533-T01-IM6	P-0019533	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446494	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837:EML4_c.3173-100:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019786-T01-IM6-84	P-0019786-T01-IM6	P-0019786	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447575	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019972-T01-IM6-85	P-0019972-T01-IM6	P-0019972	Cancer of Unknown Primary	PDC	ALK-EML4	EML4	in-frame	three_prime	29448101	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020124-T01-IM6-86	P-0020124-T01-IM6	P-0020124	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447955	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020138-T01-IM6-87	P-0020138-T01-IM6	P-0020138	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447907	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002613-T01-IM3-88	P-0002613-T01-IM3	P-0002613	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447199	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020408-T01-IM6-89	P-0020408-T01-IM6	P-0020408	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446779	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1227:EML4_c.3173-385:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003206-T01-IM5-90	P-0003206-T01-IM5	P-0003206	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29448365	19	1045	False	1045	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0003297-T01-IM5-91	P-0003297-T01-IM5	P-0003297	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448259	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020584-T01-IM6-92	P-0020584-T01-IM6	P-0020584	Renal Cell Carcinoma	URCC	ALK-EML4	EML4	in-frame	three_prime	29447048	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exon 20-29) : c.209-4846:EML4_c.3173-654:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020780-T01-IM6-93	P-0020780-T01-IM6	P-0020780	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447084	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1 - 13 fused to ALK exons 20 - 29): c.1489+1278:EML4_c.3173-690:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004218-T01-IM5-94	P-0004218-T01-IM5	P-0004218	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447087	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156:EML4_c.3173-693:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0021257-T01-IM6-95	P-0021257-T01-IM6	P-0021257	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447777	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-17 fused with ALK exons 20-29): c.1883-596:EML4_c.3172+550:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003323-T01-IM5-96	P-0003323-T01-IM5	P-0003323	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447530	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021495-T01-IM6-97	P-0021495-T01-IM6	P-0021495	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448313	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+2355:EML4_c.3172+14:ALK inv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003383-T01-IM5-98	P-0003383-T01-IM5	P-0003383	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447572	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003446-T01-IM5-99	P-0003446-T01-IM5	P-0003446	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446534	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021693-T01-IM6-100	P-0021693-T01-IM6	P-0021693	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447002	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-14 fused with ALK exons 20-29): c.1316-1509:EML4_c.3173-608:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022101-T01-IM6-101	P-0022101-T01-IM6	P-0022101	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447759	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon18 fused with ALK exon 20): c.2057-510:EML4_c.2057-510:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022138-T01-IM6-102	P-0022138-T01-IM6	P-0022138	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29446584	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2333:EML4_c.3173-190:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0022398-T01-IM6-103	P-0022398-T01-IM6	P-0022398	Non-Small Cell Lung Cancer	NSCLCPD	ALK-EML4	EML4	in-frame	three_prime	29448035	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-7853:EML4_c.3172+292:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003913-T01-IM3-104	P-0003913-T01-IM3	P-0003913	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447211	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000732-T01-IM3-105	P-0000732-T01-IM3	P-0000732	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447537	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0022926-T01-IM6-106	P-0022926-T01-IM6	P-0022926	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446558	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+983:EML4_c.3173-164:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0000474-T01-IM3-107	P-0000474-T01-IM3	P-0000474	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447002	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023368-T01-IM6-108	P-0023368-T01-IM6	P-0023368	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446713	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain): c.667+7004:EML4_c.3173-319:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0023962-T01-IM6-109	P-0023962-T01-IM6	P-0023962	Cancer of Unknown Primary	ADNOS	ALK-EML4	EML4	in-frame	three_prime	29448820	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:EML4_c.3068-389:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024136-T01-IM6-110	P-0024136-T01-IM6	P-0024136	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448266	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-4498:EML4_c.3172+61:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0005305-T01-IM5-111	P-0005305-T01-IM5	P-0005305	Non-Small Cell Lung Cancer	SARCL	ALK-EML4	EML4	in-frame	three_prime	29447764	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757:EML4_c.3172+563:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024907-T01-IM6-112	P-0024907-T01-IM6	P-0024907	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446877	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29): c.209-990:EML4_c.3173-483:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0000557-T01-IM3-113	P-0000557-T01-IM3	P-0000557	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446543	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0026650-T02-IM6-114	P-0026650-T02-IM6	P-0026650	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	five_prime	29447451	19	1058	True	1057	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKinv | Protein Fusion: in frame {ALK:EML4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026657-T01-IM6-115	P-0026657-T01-IM6	P-0026657	Small Cell Lung Cancer	CSCLC	ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0026657-T03-IM6-116	P-0026657-T03-IM6	P-0026657	Small Cell Lung Cancer	SCLC	ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027119-T01-IM6-117	P-0027119-T01-IM6	P-0027119	Thyroid Cancer	THPD	ALK-EML4	EML4	in-frame	three_prime	29446711	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused to ALK exons 20-29): c.1490-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027119-T03-IM6-118	P-0027119-T03-IM6	P-0027119	Thyroid Cancer	THPA	ALK-EML4	EML4	in-frame	three_prime	29446711	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027853-T01-IM6-119	P-0027853-T01-IM6	P-0027853	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447734	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-157:EML4_c.3172+593:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0005321-T01-IM5-120	P-0005321-T01-IM5	P-0005321	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448250	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 13 fused with ALK exon 20) : c.2068+156:EML4_c.3173-693:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028287-T01-IM6-121	P-0028287-T01-IM6	P-0028287	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446661	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-2623:EML4_c.3173-267:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028339-T01-IM6-122	P-0028339-T01-IM6	P-0028339	Salivary Gland Cancer	OSACA	ALK-EML4	EML4	in-frame	three_prime	29447680	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+1245:EML4_c.3172+647:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028541-T01-IM6-123	P-0028541-T01-IM6	P-0028541	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028541-T02-IM6-124	P-0028541-T02-IM6	P-0028541	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028541-T03-IM6-125	P-0028541-T03-IM6	P-0028541	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028745-T01-IM6-126	P-0028745-T01-IM6	P-0028745	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447509	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028901-T01-IM6-127	P-0028901-T01-IM6	P-0028901	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446607	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-844:EML4_c.3173-213:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029174-T01-IM6-128	P-0029174-T01-IM6	P-0029174	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446656	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-19 fused with ALK exons 20-29): c.2242+168:EML4_c.3173-262:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029400-T01-IM6-129	P-0029400-T01-IM6	P-0029400	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447668	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1315+641:EML4_c.3172+659:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029550-T01-IM6-130	P-0029550-T01-IM6	P-0029550	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447863	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029750-T01-IM6-131	P-0029750-T01-IM6	P-0029750	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446660	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_c.3173-266:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029971-T01-IM6-132	P-0029971-T01-IM6	P-0029971	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447807	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-119:EML4_c.3172+520:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030590-T01-IM6-133	P-0030590-T01-IM6	P-0030590	Cancer of Unknown Primary	CUP	ALK-EML4	EML4	in-frame	three_prime	29447704	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.494-5628:EML4_c.3173-350:ALKinv. | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031406-T01-IM6-134	P-0031406-T01-IM6	P-0031406	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446425	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 with ALK exons 20-29): c.668-2889:EML4_c.3173-31:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031827-T01-IM6-135	P-0031827-T01-IM6	P-0031827	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446700	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-673:EML4_c.3173-306:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031827-T02-IM6-136	P-0031827-T02-IM6	P-0031827	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446646	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032170-T01-IM6-137	P-0032170-T01-IM6	P-0032170	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447695	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1822:EML4_c.3172+632:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032518-T01-IM6-138	P-0032518-T01-IM6	P-0032518	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29448359	19	1047	False	1047	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+2607:EML4_c.3140:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0032683-T01-IM6-139	P-0032683-T01-IM6	P-0032683	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447763	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032862-T01-IM6-140	P-0032862-T01-IM6	P-0032862	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447488	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion(EML4 exon 11 fused to ALK exon20): c.1489+671:EML4_c.3172+839:ALK | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006313-T01-IM5-141	P-0006313-T01-IM5	P-0006313	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447966	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063)-ALK (NM_004304) Fusion (EML4 exons 1 to 18 fused with exons 20 to 29 of ALK) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0034450-T01-IM6-142	P-0034450-T01-IM6	P-0034450	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29446354	20	1071	False	1071	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034569-T02-IM6-143	P-0034569-T02-IM6	P-0034569	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446514	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0006370-T01-IM5-144	P-0006370-T01-IM5	P-0006370	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448124	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0001532-T01-IM3-145	P-0001532-T01-IM3	P-0001532	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447553	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035121-T01-IM6-146	P-0035121-T01-IM6	P-0035121	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447383	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4857:EML4_c.3172+944:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036106-T01-IM6-147	P-0036106-T01-IM6	P-0036106	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446930	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063)-ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036247-T01-IM6-148	P-0036247-T01-IM6	P-0036247	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447857	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006950-T01-IM5-149	P-000695 0-T01-IM5	P-000695 0	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447384	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1489+2302:EML4_c.3172+943:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006950-T02-IM6-150	P-000695 0-T02-IM6	P-000695 0	Non-Small Cell Lung Cancer	LUNE	ALK-EML4	EML4	in-frame	three_prime	29447384	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons20-29) : c.1489+2302:EML4_c.3172+943:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007125-T01-IM5-151	P-0007125-T01-IM5	P-0007125	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29446597	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328:EML4_c.3173-203:ALKinv | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0007211-T01-IM5-152	P-0007211-T01-IM5	P-0007211	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447489	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165:EML4_c.3172+838:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007705-T01-IM5-153	P-0007705-T01-IM5	P-0007705	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29447417	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1594:EML4_c.3172+910:ALKinv | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0001811-T01-IM3-154	P-0001811-T01-IM3	P-0001811	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29448205	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666:EML4_c.3172+122:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008332-T01-IM5-155	P-0008332-T01-IM5	P-0008332	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	five_prime	29446912	19	1058	True	1057	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) rearrangement : c.1489+828:EML4_c.3173-518:ALKdup | Antisense fusion	NA
EVT-P-0009102-T01-IM5-156	P-0009102-T01-IM5	P-0009102	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446513	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009301-T01-IM5-157	P-0009301-T01-IM5	P-0009301	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446651	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.668-6693:EML4_c.3173-257:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009301-T02-IM6-158	P-0009301-T02-IM6	P-0009301	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446651	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009339-T01-IM5-159	P-000933 9-T01-IM5	P-000933 9	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447889	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009339-T02-IM5-160	P-000933 9-T02-IM5	P-000933 9	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447889	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009343-T01-IM5-161	P-0009343-T01-IM5	P-0009343	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446521	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-1093:EML4_c.3173-127:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009353-T01-IM5-162	P-0009353-T01-IM5	P-0009353	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446465	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.667+2407:EML4_c.3173-71:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002024-T01-IM3-163	P-0002024-T01-IM3	P-0002024	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447133	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) inversion: c.2242+31:EML4_c.3173-739:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0002217-T02-IM3-164	P-0002217-T02-IM3	P-0002217	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29446594	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0010041-T01-IM5-165	P-0010041-T01-IM5	P-0010041	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447942	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fuses with ALK exons 20-29) : c.701-6977:EML4_c.3172+385:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0002266-T01-IM3-166	P-0002266-T01-IM3	P-0002266	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447283	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010177-T01-IM5-167	P-0010177-T01-IM5	P-0010177	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	in-frame	three_prime	29447358	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0010978-T01-IM5-168	P-0010978-T01-IM5	P-0010978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	out-of-frame	three_prime	29446414	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv | Protein fusion: out of frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010978-T02-IM5-169	P-0010978-T02-IM5	P-0010978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	out-of-frame	three_prime	29446414	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv | Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0026407-T01-IM6-170	P-0026407-T01-IM6	P-0026407	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	EML4	unknown	three_prime	29446392	20	1059	False	1059	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1981-2450:EML4_c.3175:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004516-T02-IM5-171	P-0004516-T02-IM5	P-0004516	Non-Small Cell Lung Cancer	LUAD	ALK-GCC2	GCC2	in-frame	three_prime	29446526	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			GCC2 (NM_181453) - ALK (NM_004304) rearrangement : c.4448-1230:GCC2_c.3173-132:ALKinv | Protein fusion: in frame (GCC2-ALK)	NA
EVT-P-0035712-T01-IM6-172	P-0035712-T01-IM6	P-0035712	Uterine Sarcoma	ULMS	ALK-H6PD	H6PD	unknown	five_prime	29462508	13	785	True	785	True	exact match	lost	0.0	False		Protein kinase domain		ALK (NM_004304) rearrangement: t(1;2)(p36.22;p23.2)(chr1:g.93141 50::chr2:g.29462508) | Antisense Fusion	NA
EVT-P-0018541-T01-IM6-174	P-0018541-T01-IM6	P-0018541	Bladder Cancer	UTUC	ALK-LRP1B	LRP1B	in-frame	five_prime	29450289	17	972	True	971	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ALK (NM_004304) - LRP1B (NM_018557) Rearrangement : c.2914+151:ALK_c.344-85984:LRP1Bdup | Protein Fusion: in frame {ALK:LRP1B}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027092-T01-IM6-176	P-0027092-T01-IM6	P-0027092	Non-Small Cell Lung Cancer	LUAD	ALK-PLEKHH2	PLEKHH2	in-frame	three_prime	29446506	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			PLEKHH2 (NM_172069) - ALK (NM_004304) rearrangement: c.502+545: PLEKHH2_c.3173-112:ALKinv | Protein Fusion: in frame {PLEKH2:ALK}	NA
EVT-P-0027363-T02-IM6-177	P-0027363-T02-IM6	P-0027363	Soft Tissue Sarcoma	IMT	ALK-RANBP2	RANBP2	in-frame	three_prime	29447364	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1556:RANBP2_c.3172+963:ALKinv | Protein Fusion: in frame {RANBP2:ALK}	NA
EVT-P-0008843-T01-IM5-178	P-0008843-T01-IM5	P-0008843	Soft Tissue Sarcoma	IMT	ALK-RANBP2	RANBP2	in-frame	three_prime	29446488	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1070:RANBP2_c.3173-94:ALKinv | Protein fusion: in frame (RANBP2-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068865-T02-IM7-179	P-0068865-T02-IM7	P-0068865	Soft Tissue Sarcoma	SARCNOS	ALK-RANBP2	RANBP2	in-frame	five_prime	29447373	19	1058	True	1057	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.3172+954:ALK_c.2602+1532:RANBP2inv | Protein Fusion: in frame {ALK:RANBP2}	NA
EVT-P-0015699-T01-IM6-181	P-0015699-T01-IM6	P-0015699	Colorectal Cancer	COAD	ALK-SPTBN1	SPTBN1	in-frame	three_prime	29447437	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			SPTBN1 (NM_003128) - ALK (NM_004304) fusion (SPTBN1 exon 7 fused to ALK exon 20): c.764-1028:SPTBN1_c.3172+890:ALKinv | Protein fusion: in frame (SPTBN1-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016193-T05-IM6-182	P-0016193-T05-IM6	P-0016193	Pancreatic Cancer	PAAD	ALK-STRN	STRN	unknown	three_prime	29448326	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRN_c.3172+1:ALKdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0016193-T02-IM6-183	P-0016193-T02-IM6	P-0016193	Pancreatic Cancer	PAAD	ALK-STRN	STRN	unknown	three_prime	29448326	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			ALK (NM_004304) - STRN (NM_003162) :c.3172+1:ALK_c.412+4919:STRNdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0016193-T01-IM6-184	P-0016193-T01-IM6	P-0016193	Pancreatic Cancer	PAAD	ALK-STRN	STRN	unknown	three_prime	29448326	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			ALK (NM_004304) - STRN (NM_003162) :c.3172+1:ALK_c.412+4919:STRNdel | Protein Fusion: mid-exon {STRN:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041438-T01-IM6-185	P-0041438-T01-IM6	P-0041438	Non-Small Cell Lung Cancer	LUAD	ALK-TTC27	TTC27	out-of-frame	five_prime	29447817	19	1058	True	1057	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ALK (NM_004304) - TTC27 (NM_017735) rearrangement: c.3172+510:ALK_c.538-4165:TTC27inv | Protein Fusion: out of frame {ALK:TTC27}	NA
EVT-P-0006617-T01-IM5-187	P-0006617-T01-IM5	P-0006617	Pancreatic Cancer	PAAD	ASAP2-ALK	ASAP2	unknown	five_prime	30142753	1	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ASAP2 (NM_003887) - ALK (NM_004304) rearrangement event : c.127-20148:ASAP2_c.667+106:ALKdel | Antisense fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062994-T02-IM7-188	P-0062994-T02-IM7	P-0062994	Soft Tissue Sarcoma	IMT	CLTC-ALK	CLTC	in-frame	three_prime	29449624	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29449624::chr17:g.57768350) | Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0051864-T01-IM6-189	P-0051864-T01-IM6	P-0051864	Non-Small Cell Lung Cancer	LUAD	CLTC-ALK	CLTC	in-frame	three_prime	29448839	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29448839::chr17:g.57770376) | Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0025388-T01-IM6-190	P-0025388-T01-IM6	P-0025388	Salivary Gland Cancer	SAAD	CTNNA1-ALK	CTNNA1	in-frame	three_prime	29446465	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			CTNNA1 (NM_001903) - ALK (NM_004304) rearrangment: t(2;5)(p23.2;q31.2)(chr2:g.29446465::chr5:g.138259019) | Protein Fusion: in frame {CTNNA1:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051068-T01-IM6-191	P-0051068-T01-IM6	P-0051068	Colorectal Cancer	COAD	DIAPH2-ALK	DIAPH2	in-frame	three_prime	29447566	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			DIAPH2 (NM_006729) - ALK (NM_004304) fusion: t(2;X)(p23.2;q21.33)(chr2:g.29447566::chrX:g.96713620) | Protein Fusion: in frame {DIAPH2:ALK}	NA
EVT-P-0005759-T02-IM6-192	P-0005759-T02-IM6	P-0005759	Melanoma	SKCM	EMILIN1-ALK	EMILIN1	in-frame	three_prime	29450741	17	939	True	939	True	exact match	retained	1.0	False	Protein kinase domain			EMILIN1 (NM_007046) - ALK (NM_004304) fusion: c.2575+44:EMILIN1_c.2816-203:ALKinv | Protein Fusion: in frame {EMILIN1:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022184-T01-IM6-193	P-0022184-T01-IM6	P-0022184	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447035	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1315+1386:EML4_c.3173-641:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069617-T01-IM7-194	P-0069617-T01-IM7	P-0069617	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448017	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013299-T02-IM6-195	P-0013299-T02-IM6	P-0013299	Breast Cancer	IDC	EML4-ALK	EML4	in-frame	three_prime	29447038	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063)-ALK (NM_004304) Fusion(EML4 exon5 fused with ALK exon20): c.667+1302:EML4_c.3173-644:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034334-T01-IM6-196	P-0034334-T01-IM6	P-0034334	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446570	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025040-T01-IM6-197	P-0025040-T01-IM6	P-0025040	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447152	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0025095-T01-IM6-198	P-0025095-T01-IM6	P-0025095	Non-Small Cell Lung Cancer	NSCLC	EML4-ALK	EML4	in-frame	three_prime	29447321	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-278:EML4_c.3173-927:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031827-T03-IM6-199	P-0031827-T03-IM6	P-0031827	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0035085-T03-IM6-200	P-0035085-T03-IM6	P-0035085	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447079	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7361:EML4_c.3173-685:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0005743-T02-IM5-201	P-0005743-T02-IM5	P-0005743	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447669	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243-181:EML4_c.3172+658:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010177-T02-IM7-202	P-0010177-T02-IM7	P-0010177	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447358	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031827-T06-IM7-203	P-0031827-T06-IM7	P-0031827	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036106-T03-IM7-204	P-0036106-T03-IM7	P-0036106	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446930	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-536:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005786-T01-IM5-205	P-0005786-T01-IM5	P-0005786	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448239	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 with ALK exons 20-29) : c.2057-186:EML4_c.3172+88:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0037404-T01-IM6-206	P-0037404-T01-IM6	P-0037404	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447872	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4963:EML4_c.3172+455:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037491-T01-IM6-207	P-0037491-T01-IM6	P-0037491	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446786	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon12 fused with ALK exon20): c.1490-1120:EML4_c.3173-392:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037668-T01-IM6-208	P-0037668-T01-IM6	P-0037668	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447631	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2742:EML4_c.3172+696:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037925-T01-IM6-209	P-0037925-T01-IM6	P-0037925	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446522	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4634:EML4_c.3173-128:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037982-T01-IM6-210	P-0037982-T01-IM6	P-0037982	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446479	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-138:EML4_c.3173-85:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038201-T03-IM6-211	P-0038201-T03-IM6	P-0038201	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447748	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+989:EML4_c.3172+579:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038772-T01-IM6-212	P-0038772-T01-IM6	P-0038772	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446687	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2341+147:EML4_c.3173-293:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014310-T02-IM6-213	P-0014310-T02-IM6	P-0014310	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447544	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+6128:EML4_c.3172+783:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017592-T04-IM6-214	P-0017592-T04-IM6	P-0017592	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447548	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-5963:EML4_c.3172+779:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040261-T01-IM6-215	P-0040261-T01-IM6	P-0040261	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447103	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1340:EML4_c.3173-709:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027119-T04-IM7-216	P-0027119-T04-IM7	P-0027119	Thyroid Cancer	THPD	EML4-ALK	EML4	in-frame	three_prime	29446711	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040553-T01-IM6-217	P-0040553-T01-IM6	P-0040553	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447055	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+30:EML4_c.3173-661:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040574-T01-IM6-218	P-0040574-T01-IM6	P-0040574	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447738	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-1589:EML4_c.3172+589:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040576-T01-IM6-219	P-0040576-T01-IM6	P-0040576	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447321	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-821:EML4_c.3173-927:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040600-T01-IM6-220	P-0040600-T01-IM6	P-0040600	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447688	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1795:EML4_c.3172+639:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040732-T02-IM6-221	P-0040732-T02-IM6	P-0040732	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446636	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1832:EML4_c.3173-242:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001766-T04-IM6-222	P-0001766-T04-IM6	P-0001766	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448767	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3962:EML4_c.3068-336:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014385-T02-IM6-223	P-0014385-T02-IM6	P-0014385	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447676	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041967-T01-IM6-224	P-0041967-T01-IM6	P-0041967	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448147	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041967-T02-IM7-225	P-0041967-T02-IM7	P-0041967	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448147	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014385-T03-IM6-226	P-0014385-T03-IM6	P-0014385	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447676	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043041-T01-IM6-227	P-0043041-T01-IM6	P-0043041	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446597	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1782:EML4_c.3173-203:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014385-T06-IM7-228	P-0014385-T06-IM7	P-0014385	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447676	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043350-T01-IM6-229	P-0043350-T01-IM6	P-0043350	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447513	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4017:EML4_c.3172+814:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043363-T01-IM6-230	P-0043363-T01-IM6	P-0043363	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	unknown	three_prime	29448101	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2254:EML4_c.3172+226:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043594-T01-IM6-231	P-0043594-T01-IM6	P-0043594	Colorectal Cancer	MACR	EML4-ALK	EML4	unknown	three_prime	29448381	19	1040	False	1040	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0043860-T02-IM6-232	P-0043860-T02-IM6	P-0043860	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446931	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-980:EML4_c.3173-537:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0007705-T02-IM6-233	P-0007705-T02-IM6	P-0007705	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	unknown	three_prime	29447417	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044042-T01-IM6-234	P-0044042-T01-IM6	P-0044042	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447727	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1681:EML4_c.3172+600:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044432-T01-IM6-235	P-0044432-T01-IM6	P-0044432	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	unknown	three_prime	29448344	19	1052	False	1052	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2057-576:EML4_c.3155:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0045222-T03-IM7-236	P-0045222-T03-IM7	P-0045222	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	unknown	three_prime	29447583	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045243-T01-IM6-237	P-0045243-T01-IM6	P-0045243	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448298	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0045341-T01-IM6-238	P-0045341-T01-IM6	P-0045341	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447920	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012913-T02-IM6-239	P-0012913-T02-IM6	P-0012913	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446965	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-4133:EML4_c.3173-571:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045912-T02-IM6-240	P-0045912-T02-IM6	P-0045912	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	out-of-frame	three_prime	29446507	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+1011:EML4_c.3173-113:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0046083-T01-IM6-241	P-0046083-T01-IM6	P-0046083	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446665	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986:EML4_c.3173-271:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046083-T02-IM7-242	P-0046083-T02-IM7	P-0046083	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446665	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986:EML4_c.3173-271:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046594-T01-IM6-243	P-0046594-T01-IM6	P-0046594	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446445	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2217:EML4_c.3173-51:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046720-T01-IM6-244	P-0046720-T01-IM6	P-0046720	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	out-of-frame	three_prime	29446669	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1641+1:EML4_c.3173-275:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0047355-T01-IM6-245	P-0047355-T01-IM6	P-0047355	Non-Small Cell Lung Cancer	LUSC	EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047355-T02-IM6-246	P-0047355-T02-IM6	P-0047355	Non-Small Cell Lung Cancer	LUSC	EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019560-T01-IM6-247	P-0019560-T01-IM6	P-0019560	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	out-of-frame	three_prime	29447952	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-16 with ALK exons 20-29): c.1968-2879:EML4_c.3172+375:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0049318-T01-IM6-248	P-0049318-T01-IM6	P-0049318	Thyroid Cancer	THHC	EML4-ALK	EML4	unknown	three_prime	29446359	20	1070	False	1070	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+2335:EML4_c.3208:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023375-T02-IM7-249	P-0023375-T02-IM7	P-0023375	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447854	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2728:EML4_c.3172+473:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049375-T02-IM6-250	P-0049375-T02-IM6	P-0049375	Non-Small Cell Lung Cancer	SARCL	EML4-ALK	EML4	in-frame	three_prime	29446732	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049511-T01-IM6-251	P-0049511-T01-IM6	P-0049511	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446991	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022184-T04-IM6-252	P-0022184-T04-IM6	P-0022184	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447035	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1386:EML4_c.3173-641:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015961-T07-IM6-253	P-0015961-T07-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAS	EML4-ALK	EML4	in-frame	three_prime	29446526	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051231-T01-IM6-254	P-0051231-T01-IM6	P-0051231	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446774	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051231-T03-IM6-255	P-0051231-T03-IM6	P-0051231	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446774	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0052035-T03-IM6-256	P-0052035-T03-IM6	P-0052035	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447656	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+79:EML4_c.3172+671:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0052297-T02-IM6-257	P-0052297-T02-IM6	P-0052297	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448255	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-161:EML4_c.3172+72:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054263-T01-IM6-258	P-0054263-T01-IM6	P-0054263	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446557	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-210:EML4_c.3173-163:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015646-T03-IM6-259	P-0015646-T03-IM6	P-0015646	Nerve Sheath Tumor	MPNST	EML4-ALK	EML4	in-frame	three_prime	29448028	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-3909:EML4_c.3172+299:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054408-T01-IM6-260	P-0054408-T01-IM6	P-0054408	Non-Small Cell Lung Cancer	LUAS	EML4-ALK	EML4	in-frame	three_prime	29446902	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2243-262:EML4_c.3173-508:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008332-T02-IM6-261	P-0008332-T02-IM6	P-0008332	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	unknown	three_prime	29446913	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828:EML4_c.3173-519:ALKdup | Antisense Fusion	NA
EVT-P-0057391-T02-IM6-262	P-0057391-T02-IM6	P-0057391	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447092	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0058599-T02-IM7-263	P-0058599-T02-IM7	P-0058599	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446415	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1345:EML4_c.3173-21:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059257-T02-IM7-264	P-0059257-T02-IM7	P-0059257	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448210	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-982:EML4_c.3172+117:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012873-T02-IM6-265	P-0012873-T02-IM6	P-0012873	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0059699-T03-IM7-266	P-0059699-T03-IM7	P-0059699	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447467	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1990:EML4_c.3172+860:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060148-T01-IM7-267	P-0060148-T01-IM7	P-0060148	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447757	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2600:EML4_c.3172+570:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0061297-T02-IM7-268	P-0061297-T02-IM7	P-0061297	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	unknown	three_prime	29448375	19	1042	False	1042	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+931:EML4_c.3124:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0061297-T03-IM7-269	P-0061297-T03-IM7	P-0061297	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	unknown	three_prime	29448375	19	1042	False	1042	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+931:EML4_c.3124:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062985-T02-IM7-270	P-0062985-T02-IM7	P-0062985	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447199	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4561:EML4_c.3173-805:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032683-T02-IM7-271	P-0032683-T02-IM7	P-0032683	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447763	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0062999-T02-IM7-272	P-0062999-T02-IM7	P-0062999	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447091	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0063585-T01-IM7-273	P-0063585-T01-IM7	P-0063585	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446884	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064342-T01-IM7-274	P-0064342-T01-IM7	P-0064342	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446534	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064716-T02-IM7-275	P-0064716-T02-IM7	P-0064716	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447739	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3669:EML4_c.3172+588:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065210-T01-IM7-276	P-0065210-T01-IM7	P-0065210	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448076	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2012:EML4_c.3172+251:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065469-T02-IM7-277	P-0065469-T02-IM7	P-0065469	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446565	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031368-T04-IM6-278	P-0031368-T04-IM6	P-0031368	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447191	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1154:EML4_c.3173-797:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0066568-T02-IM7-279	P-0066568-T02-IM7	P-0066568	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448072	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+3428:EML4_c.3172+255:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067048-T02-IM7-280	P-0067048-T02-IM7	P-0067048	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447391	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2216:EML4_c.3172+936:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067069-T01-IM7-281	P-0067069-T01-IM7	P-0067069	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448169	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2305:EML4_c.3172+158:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0067093-T02-IM7-282	P-0067093-T02-IM7	P-0067093	Renal Cell Carcinoma	URCC	EML4-ALK	EML4	in-frame	three_prime	29447303	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+1508:EML4_c.3173-909:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012873-T03-IM7-283	P-0012873-T03-IM7	P-0012873	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0068467-T02-IM7-284	P-0068467-T02-IM7	P-0068467	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29448295	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1566:EML4_c.3172+32:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031824-T01-IM6-285	P-0031824-T01-IM6	P-0031824	Pancreatic Cancer	PAAD	EML4-ALK	EML4	in-frame	three_prime	29450557	17	939	True	939	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+2971:EML4_c.2816-19:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069396-T02-IM7-286	P-0069396-T02-IM7	P-0069396	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446944	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069535-T01-IM7-287	P-0069535-T01-IM7	P-0069535	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29447764	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049375-T01-IM6-288	P-0049375-T01-IM6	P-0049375	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29446732	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044526-T02-IM6-289	P-0044526-T02-IM6	P-0044526	Soft Tissue Sarcoma	IMT	FN1-ALK	FN1	in-frame	three_prime	29448533	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			FN1 (NM_212482) - ALK (NM_004304) fusion: c.1819+332:FN1_c.3068-102:ALKdel | Protein Fusion: in frame {FN1:ALK}	NA
EVT-P-0030808-T01-IM6-290	P-0030808-T01-IM6	P-0030808	Non-Small Cell Lung Cancer	LUAD	HMBOX1-ALK	HMBOX1	unknown	three_prime	29447713	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			HMBOX1 (NM_001135726) ALK (NM_004304) Rearrangement : t(2,8)(p23.3,p21.1)(chr2:g.29447713::chr8:g.28904937) | Protein Fusion: mid-exon {HMBOX1:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008764-T01-IM5-291	P-0008764-T01-IM5	P-0008764	Non-Small Cell Lung Cancer	LUAD	KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	1058	True	exact match	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750) | Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0062804-T03-IM7-292	P-0062804-T03-IM7	P-0062804	Soft Tissue Sarcoma	IMT	KIF5B-ALK	KIF5B	in-frame	three_prime	29447791	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447791::chr10:g.32304833) | Protein Fusion: in frame {KIF5B:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017924-T01-IM6-293	P-0017924-T01-IM6	P-0017924	Non-Small Cell Lung Cancer	LUAD	KIF5B-ALK	KIF5B	in-frame	three_prime	29447945	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447945::chr10:g.32305310) | Protein Fusion: in frame {KIF5B:ALK}	NA
EVT-P-0005711-T02-IM6-294	P-0005711-T02-IM6	P-0005711	Non-Small Cell Lung Cancer	NSCLCPD	KIF5B-ALK	KIF5B	in-frame	three_prime	29449548	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29449548::chr10:g.32304765) | Protein fusion: in frame (KIF5B-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008764-T02-IM6-295	P-0008764-T02-IM6	P-0008764	Non-Small Cell Lung Cancer	LUAD	KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	1058	True	exact match	retained	1.0	False	Protein kinase domain			KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.:32305750) | Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0067512-T01-IM7-297	P-0067512-T01-IM7	P-0067512	Glioma	GB	NOTCH1-ALK	NOTCH1	in-frame	three_prime	29449152	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			NOTCH1 (NM_017617) - ALK (NM_004304) fusion: t(2;9)(p23.2;q34.3)(chr2:g.29449152::chr9:g.139416748) | Protein Fusion: in frame {NOTCH1:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034044-T01-IM6-298	P-0034044-T01-IM6	P-0034044	Uterine Sarcoma	USTUMP	NRP2-ALK	NRP2	in-frame	three_prime	29448550	19	1023	True	1023	True	exact match	retained	1.0	False	Protein kinase domain			NRP2 (NM_201266) - ALK (NM_004304) rearrangement: c.1291+575_c.3068-119inv | Protein Fusion: in frame {NRP2:ALK}	NA
EVT-P-0055667-T02-IM6-299	P-0055667-T02-IM6	P-0055667	Non-Small Cell Lung Cancer	LNET	PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	1061	True	exact match	retained	1.0	False	Protein kinase domain			PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) | Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0055667-T04-IM6-300	P-0055667-T04-IM6	P-0055667	Cancer of Unknown Primary	NETNOS	PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	1061	True	exact match	retained	1.0	False	Protein kinase domain			PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) | Protein Fusion: mid-exon {PICALM:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055667-T06-IM7-301	P-0055667-T06-IM7	P-0055667	Non-Small Cell Lung Cancer	LUAD	PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	1061	True	exact match	retained	1.0	False	Protein kinase domain			PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) | Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0054253-T01-IM6-302	P-0054253-T01-IM6	P-0054253	Non-Small Cell Lung Cancer	LUAD	PLEKHA7-ALK	PLEKHA7	unknown	three_prime	29448396	19	1035	False	1035	True	exact match	retained	1.0	False	Protein kinase domain			PLEKHA7 (NM_175058) - ALK (NM_004304) fusion: t(2;11)(p23.2;p15.1)(chr2:g.29448396::chr11:g.16813809) | Protein Fusion: mid-exon {PLEKHA7:ALK}	NA
EVT-P-0059884-T01-IM7-303	P-0059884-T01-IM7	P-0059884	Ovarian Cancer	HGSOC	PPP1CB-ALK	PPP1CB	in-frame	three_prime	29447441	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			PPP1CB (NM_206876) - ALK (NM_004304) fusion: c.52+5708:PPP1CB_c.3172+886:ALKinv | Protein Fusion: in frame {PPP1CB:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048893-T01-IM6-307	P-0048893-T01-IM6	P-0048893	Colorectal Cancer	COAD	SPTBN1-ALK	SPTBN1	unknown	three_prime	29448346	19	1051	False	1051	True	exact match	retained	1.0	False	Protein kinase domain			SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.763+1517:SPTBN1_c.3153:ALKinv | Protein Fusion: mid-exon {SPTBN1:ALK}	NA
EVT-P-0045724-T01-IM6-308	P-0045724-T01-IM6	P-0045724	Colorectal Cancer	COAD	SPTBN1-ALK	SPTBN1	in-frame	three_prime	29446796	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-456:SPTBN1_c.3173-402:ALKinv | Protein Fusion: in frame {SPTBN1:ALK}	NA
EVT-P-0053963-T02-IM6-309	P-0053963-T02-IM6	P-0053963	Non-Small Cell Lung Cancer	LUAD	SPTBN1-ALK	SPTBN1	unknown	three_prime	29448345	19	1052	False	1052	True	exact match	retained	1.0	False	Protein kinase domain			SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-1123:SPTBN1_c.3154:ALKinv | Protein Fusion: mid-exon {SPTBN1:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042596-T02-IM6-310	P-0042596-T02-IM6	P-0042596	Non-Small Cell Lung Cancer	LUAD	SQSTM1-ALK	SQSTM1	in-frame	three_prime	29446511	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t(2;5)(p23.2;q35.3)(chr2:g.29446511::chr5:g.179254603) | Protein Fusion: in frame {SQSTM1:ALK}	NA
EVT-P-0042596-T01-IM6-311	P-0042596-T01-IM6	P-0042596	Non-Small Cell Lung Cancer	LUAD	SQSTM1-ALK	SQSTM1	in-frame	three_prime	29446511	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t(2;5)(p23.2;q35.3)(chr2:g.29446511::chr5:g.179254603) | Protein Fusion: in frame {SQSTM1:ALK}	NA
EVT-P-0048102-T02-IM6-312	P-0048102-T02-IM6	P-0048102	Non-Small Cell Lung Cancer	LUAD	STRN-ALK	STRN	in-frame	three_prime	29447056	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			STRN (NM_003162) - ALK (NM_004304) fusion: c.412+2761:STRN_c.3173-662:ALKdel | Protein Fusion: in frame {STRN:ALK}	NA

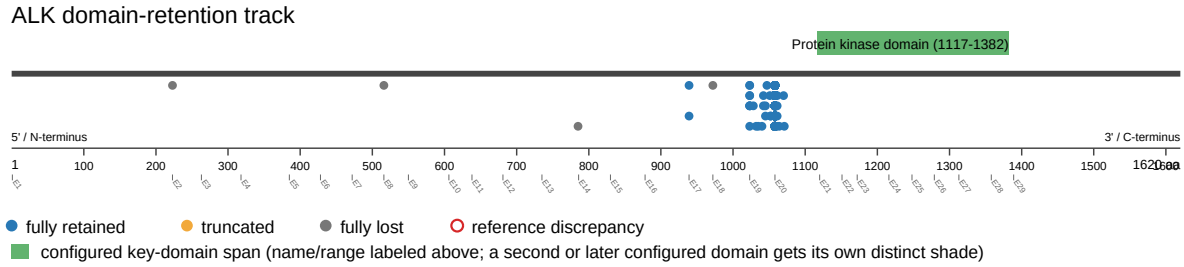
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048102-T03-IM6-313	P-0048102-T03-IM6	P-0048102	Non-Small Cell Lung Cancer	LUAD	STRN-ALK	STRN	in-frame	three_prime	29447056	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			STRN (NM_003162) - ALK (NM_004304) fusion: c.412+2761:STRN_c.3173-662:ALKdel | Protein Fusion: in frame {STRN:ALK}	NA
EVT-P-0033066-T01-IM6-314	P-0033066-T01-IM6	P-0033066	Colorectal Cancer	COAD	STRN-ALK	STRN	unknown	three_prime	29448365	19	1045	False	1045	True	exact match	retained	1.0	False	Protein kinase domain			STRN (NM_003162) - ALK (NM_004304) fusion(STRN exons 1-3 fused to ALK exons 19-29): c.413-1084:STRN_c.3134:ALKdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0053901-T01-IM6-315	P-0053901-T01-IM6	P-0053901	Colorectal Cancer	COAD	STRN-ALK	STRN	unknown	three_prime	29446375	20	1064	False	1064	True	exact match	retained	1.0	False	Protein kinase domain			STRN (NM_003162) - ALK (NM_004304) fusion: c.339-2565:STRN_c.3192:ALKdel | Protein Fusion: mid-exon {STRN:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0056097-T01-IM6-316	P-0056097-T01-IM6	P-0056097	Non-Small Cell Lung Cancer	LUAD	TFG-ALK	TFG	in-frame	three_prime	29447896	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			TFG (NM_001195478) - ALK (NM_004304) rearrangement: t(2;3)(p23.2;q12.2)(chr2:g.29447896::chr3:g.100456163) | Protein Fusion: in frame {TFG:ALK}	NA
EVT-P-0045363-T01-IM6-318	P-0045363-T01-IM6	P-0045363	Mesothelioma	PEMESO	TPM1-ALK	TPM1	in-frame	three_prime	29446514	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			TPM1 (NM_001018020) - ALK (NM_004304) fusion: t(2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833) | Protein Fusion: in frame {TPM1:ALK}	NA
EVT-P-0045363-T02-IM6-319	P-0045363-T02-IM6	P-0045363	Mesothelioma	PEMESO	TPM1-ALK	TPM1	in-frame	three_prime	29446514	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			TPM1 (NM_001018020) - ALK (NM_004304) fusion: t(2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833) | Protein Fusion: in frame {TPM1:ALK}	NA

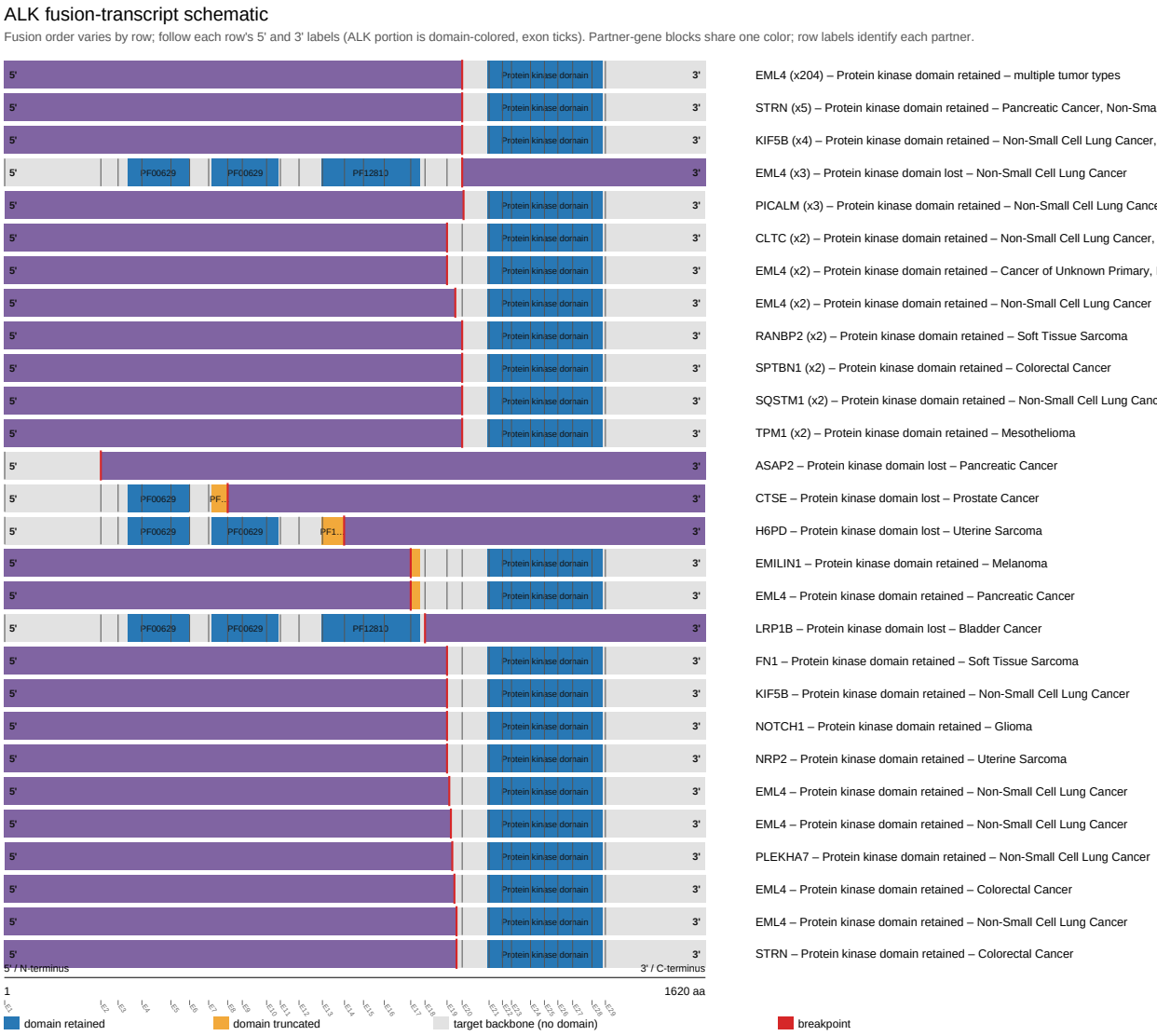
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053383-T01-IM6-320	P-0053383-T01-IM6	P-0053383	Soft Tissue Sarcoma	IMT	TRAF3-ALK	TRAF3	in-frame	three_prime	29447688	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			TRAF3 (NM_003300) - ALK (NM_004304) fusion: t(2;14)(p23.2;q32.32)(chr2:g.29447688::chr14:g.103370282) | Protein Fusion: in frame {TRAF3:ALK}	NA
EVT-P-0001737-T01-IM3-321	P-0001737-T01-IM3	P-0001737	Renal Cell Carcinoma	URCC	VCL-ALK	VCL	in-frame	three_prime	29447390	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			VCL (NM_014000) - ALK (NM_004304) Translocation : t(10;2)(q22.2;p23.2)(chr10:c.2435-486::chr2:c.3172+937) | Protein fusion: in frame (VCL-ALK)	NA
EVT-P-0045566-T01-IM6-322	P-0045566-T01-IM6	P-0045566	Non-Small Cell Lung Cancer	LUAD	ZFPM2-ALK	ZFPM2	in-frame	three_prime	29447902	20	1058	True	1058	True	exact match	retained	1.0	False	Protein kinase domain			ZFPM2 (NM_012082) - ALK (NM_004304) fusion: t(2;8)(p23.2;q23.1)(chr2:g.29447902::chr8:g.106384092) | Protein Fusion: in frame {ZFPM2:ALK}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic

